$\min f(x) \quad \text{s.t. } h(x) = 0 \quad (P)$

$L(x, \lambda) = f(x) + \lambda h(x)$

$g(\lambda) = \min_x L(x, \lambda)$

$\max g(\lambda) \quad \text{s.t. } \lambda \geq 0 \quad (D)$

Grad Des $\longrightarrow$ Dual Aes

$\downarrow$ $L + \frac{1}{2}\rho h^2$ 增广

ALM

$\downarrow$ 当 $\min f + g$ 时

ADMM

Dual Ascent Method

$$x^{k+1} = \operatorname{argmin} L(x, \lambda^k)$$

$$\lambda^{k+1} = \lambda^k + \alpha \frac{\partial}{\partial \lambda} L(x^{k+1}, \lambda) = \lambda^k + \alpha h(x^{k+1})$$

Augment Lagrangian $\quad AL = f + \lambda h + \frac{1}{2}\rho h^2$

ALM (Augment Lagrangian Method)

$$x_{k+1} = \operatorname{argmin} AL(x, \lambda_k)$$

$$\lambda_{k+1} = \lambda_k + \rho h(x_{k+1})$$

α 取 ρ 即可

ADMM

$\min f(x) + g(z) \quad \text{s.t. } Ax + Bz = c$

$L_a = f + g + \langle \lambda, Ax + Bz - c \rangle + \frac{1}{2}\rho \| Ax + Bz - c \|^2$

$x_k = \operatorname{argmin} L(x, z_{k-1}, \lambda_{k-1})$

$z_k = \operatorname{argmin} L(x_k, z, \lambda_{k-1})$

$\lambda_k = \lambda_{k-1} + \rho(Ax_k + Bz_k - c)$

收敛性：若 A1. A2 满足，点收敛值也收敛

A1. f, g 闭凸

A2. 未增广 L_0 有鞍点